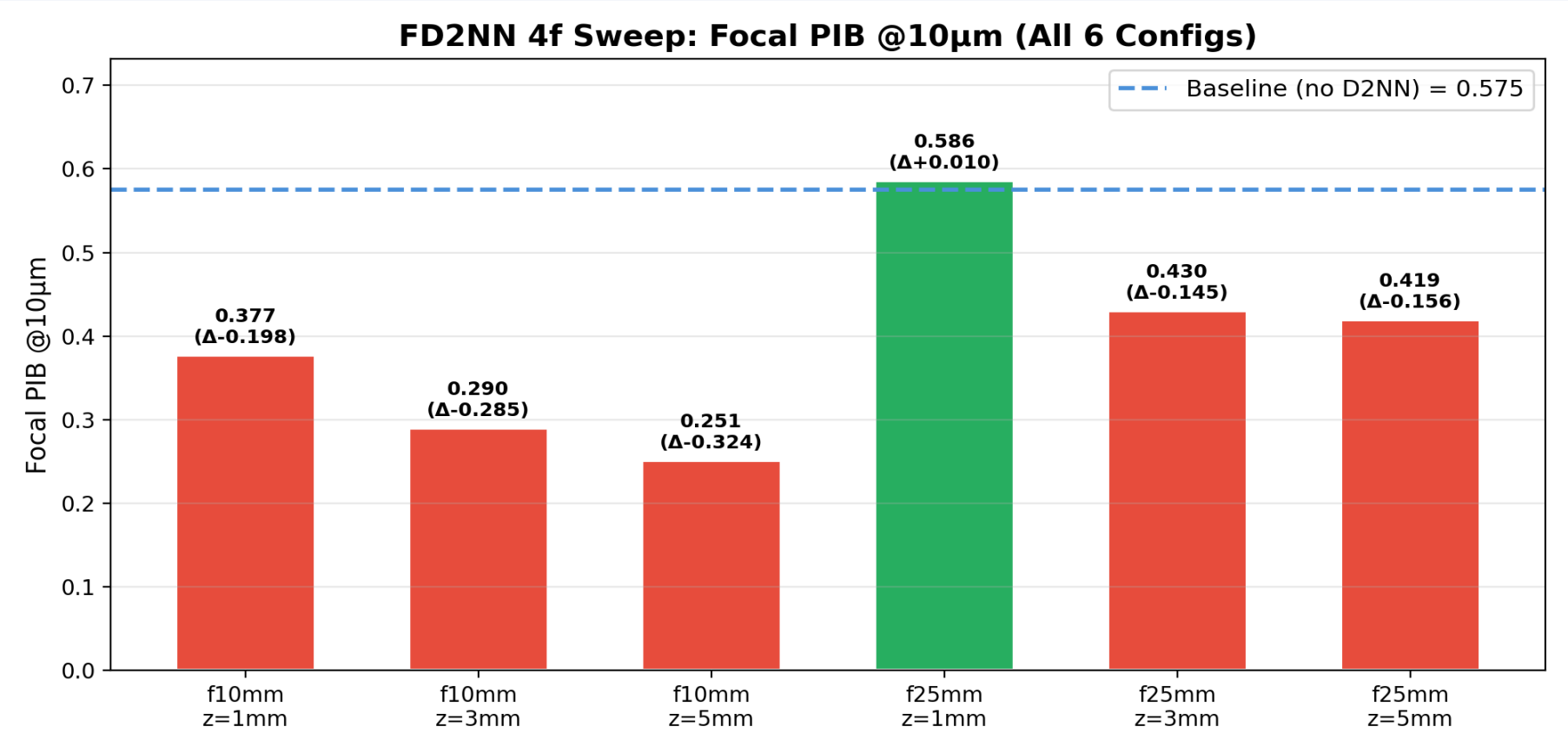
FD2NN 4f Multi-layer Sweep 실험 결과 보고

Architecture: `Input → Lens(f) → [Mask₁→ASM(z)→...→Mask₅] → Lens(f) → Focal Lens(6.5mm) → PIB`

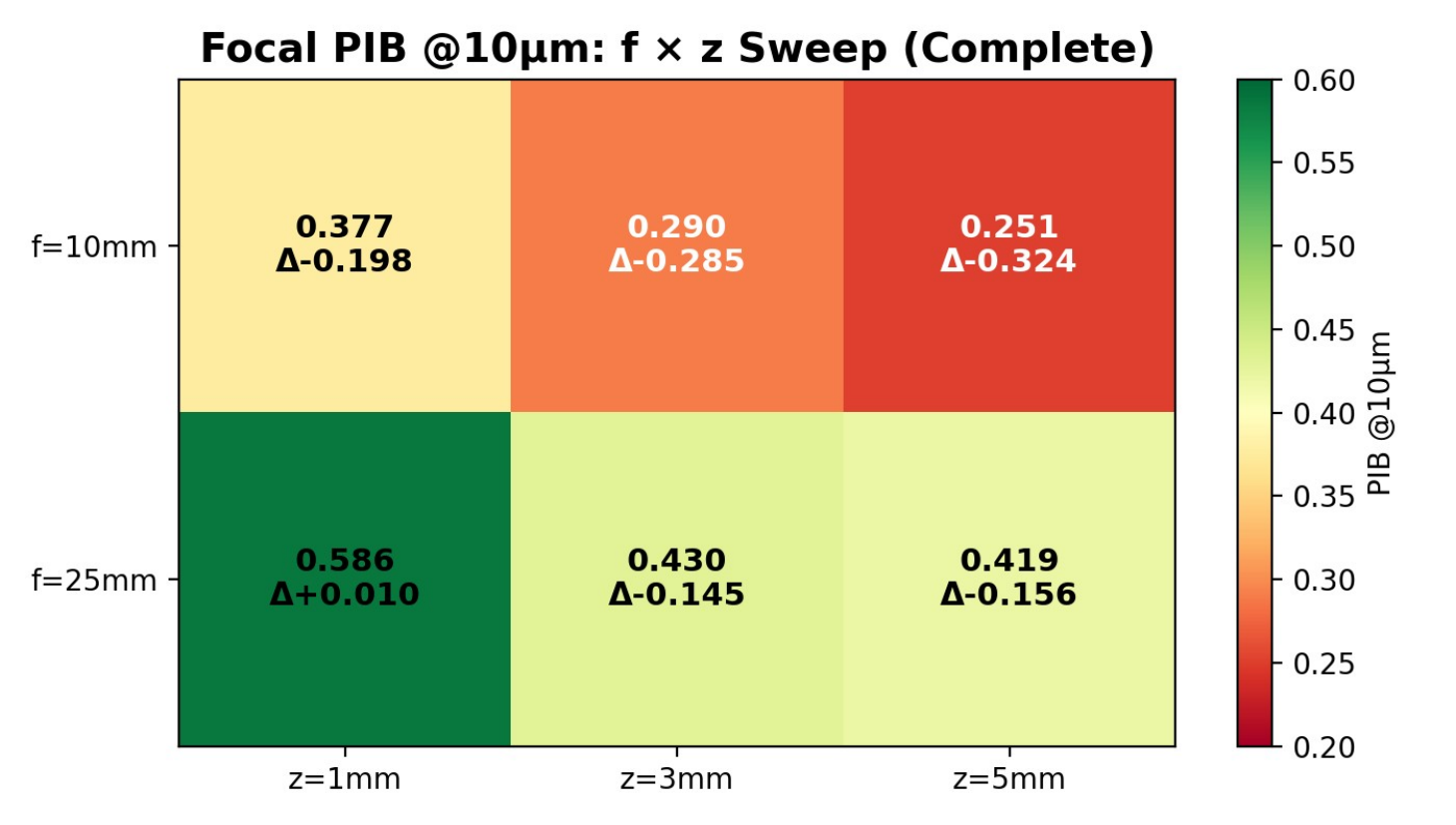
Sweep: f ∈ {10, 25}mm × z ∈ {1, 3, 5}mm = 6 configs | Loss: focal_raw_received_power

2026-04-06 | D2NN Lab

Focal PIB @10μm — 전체 결과



f × z Sweep Heatmap



정량적 결과 요약

Config	f (mm)	z (mm)	dx_f (μm)	z/z_R	fPIB@10μm	Δ PIB	CO	TP
f10mm_z1mm	10	1	7.6	0.67	0.377	-0.198	0.204	1.000
f10mm_z3mm	10	3	7.6	2.0	0.290	-0.285	0.190	1.000
f10mm_z5mm	10	5	7.6	3.3	0.251	-0.324	0.174	1.000
f25mm_z1mm	25	1	18.9	0.028	**0.586**	+0.010	**0.249**	1.000
f25mm_z3mm	25	3	18.9	0.083	0.430	-0.145	0.220	1.000
f25mm_z5mm	25	5	18.9	0.139	0.419	-0.156	0.218	1.000

Baseline (no D2NN): fPIB@10μm = 0.575, CO = 0.268

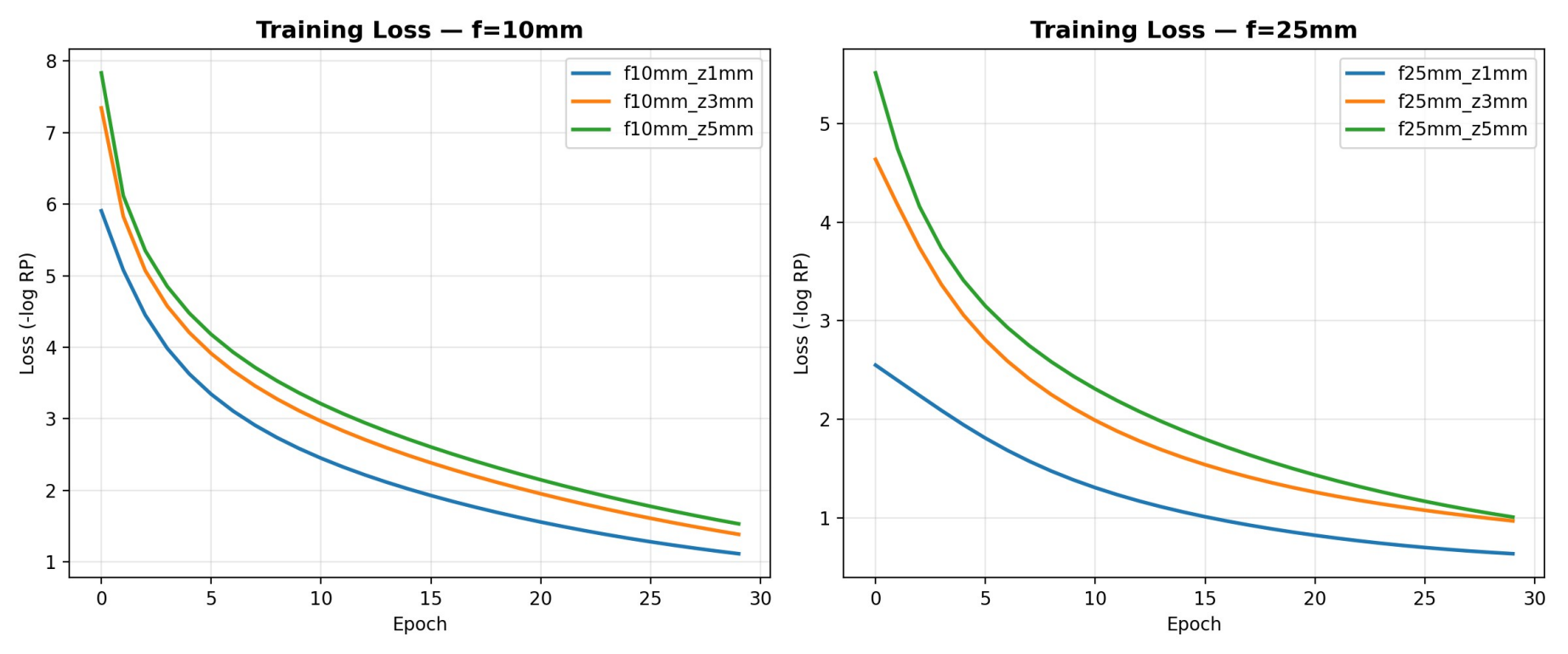
f=10mm: 모두 baseline 미달

z/z_R > 0.67 → defocus 심각
학습이 PIB 0.3%→37.7% 로 복구했지만
baseline(57.5%) 까지 못 도달

f=25mm: baseline 과 동등

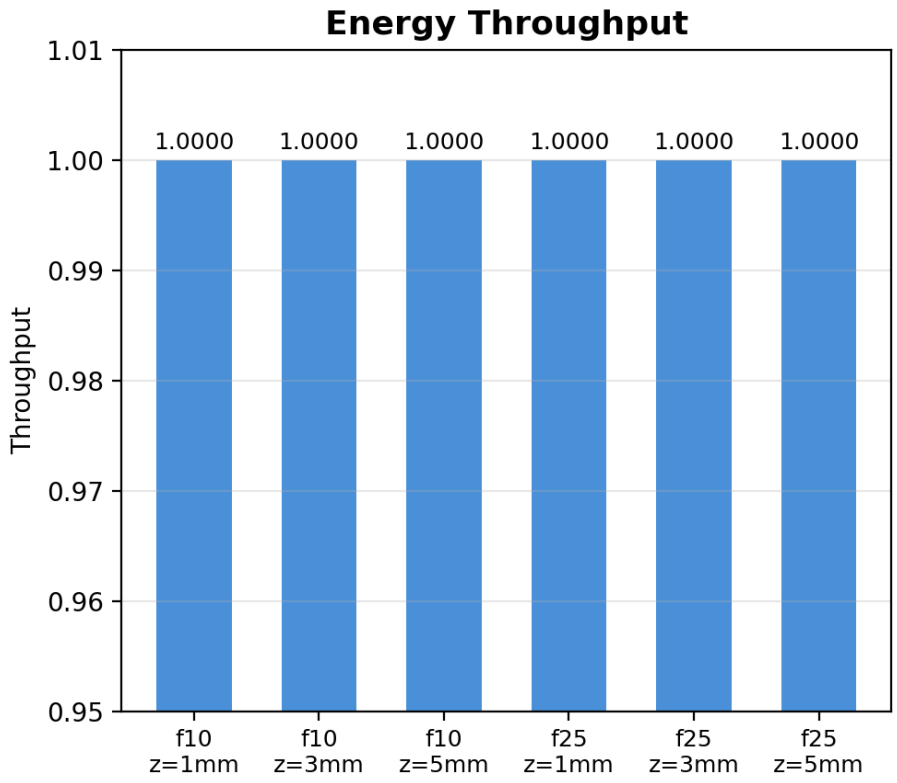
z/z_R < 0.1 → defocus 없음
baseline 에서 시작하지만
회절 mixing 부족 → Δ=+0.01

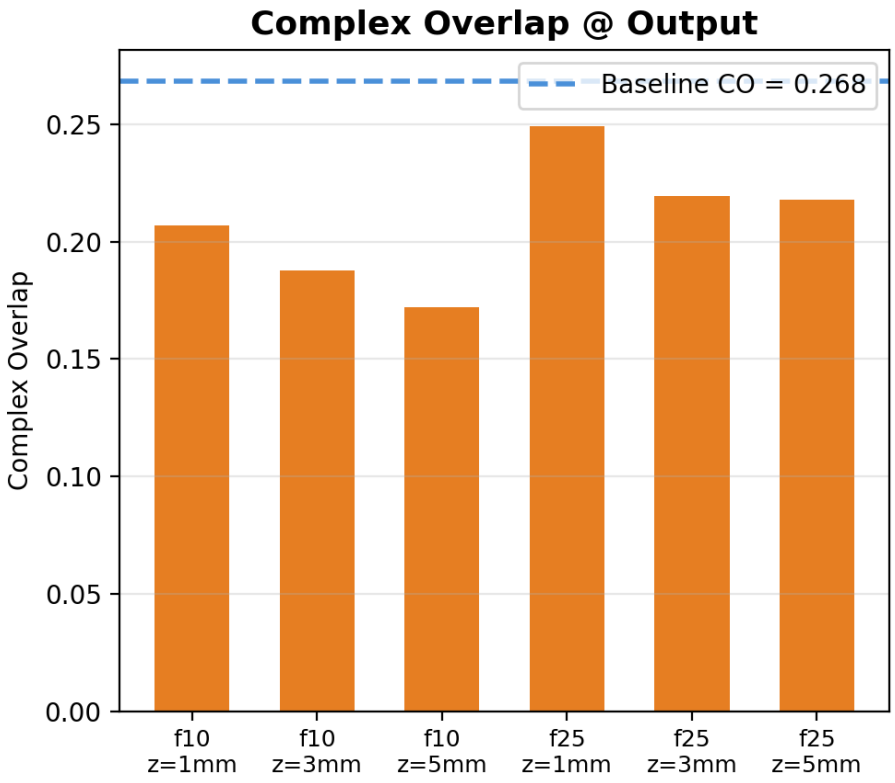
Training Loss Curves



f=10mm: loss > 1.0 (exp(-1.0) ≈ 37% RP). z↑ 일수록 수렴이 느리고 final loss 높음 .
f=25mm: loss < 1.0 달성 (exp(-0.63) ≈ 53% RP). 더 효율적인 에너지 집중 .

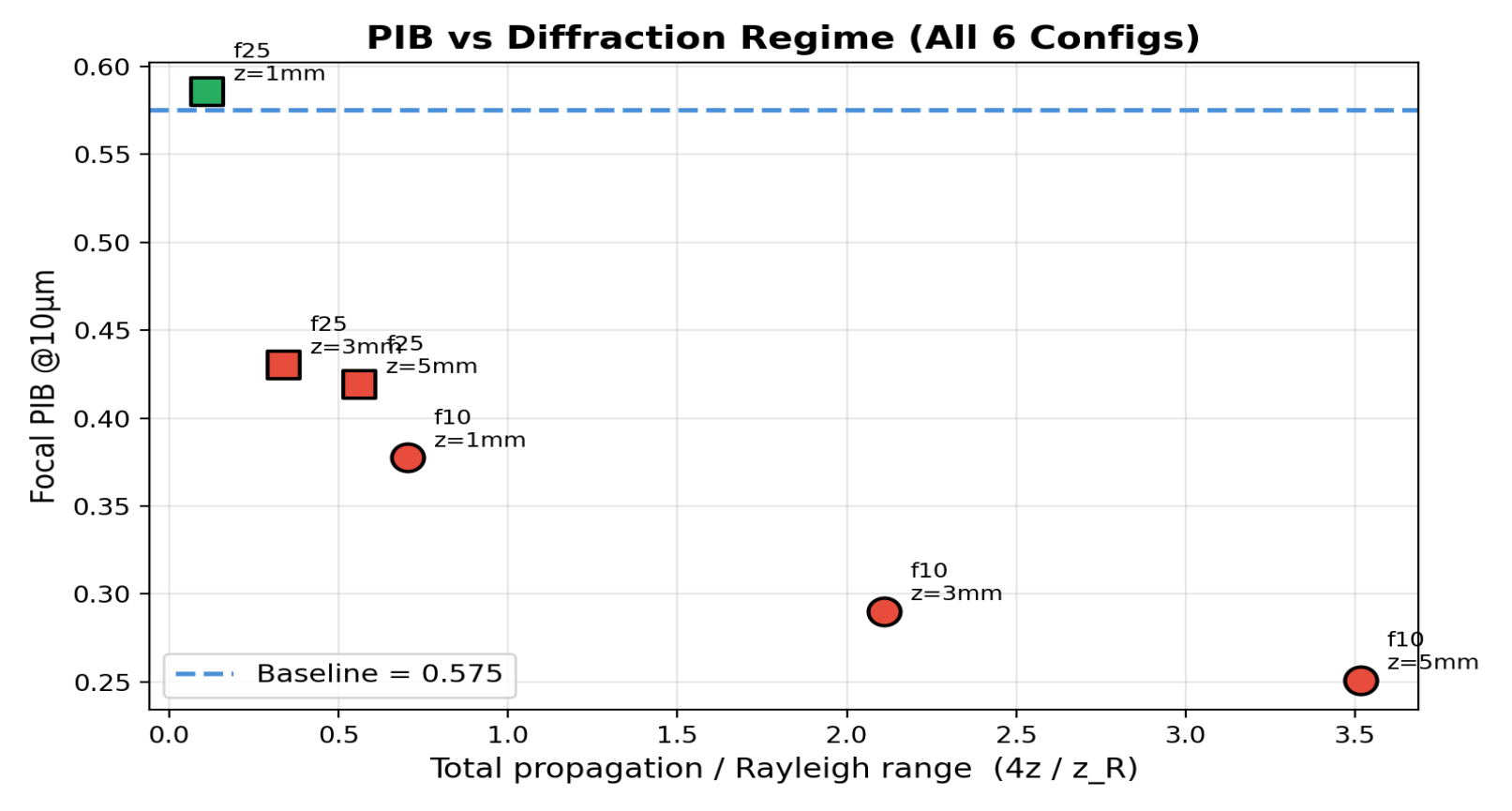
Throughput & Complex Overlap



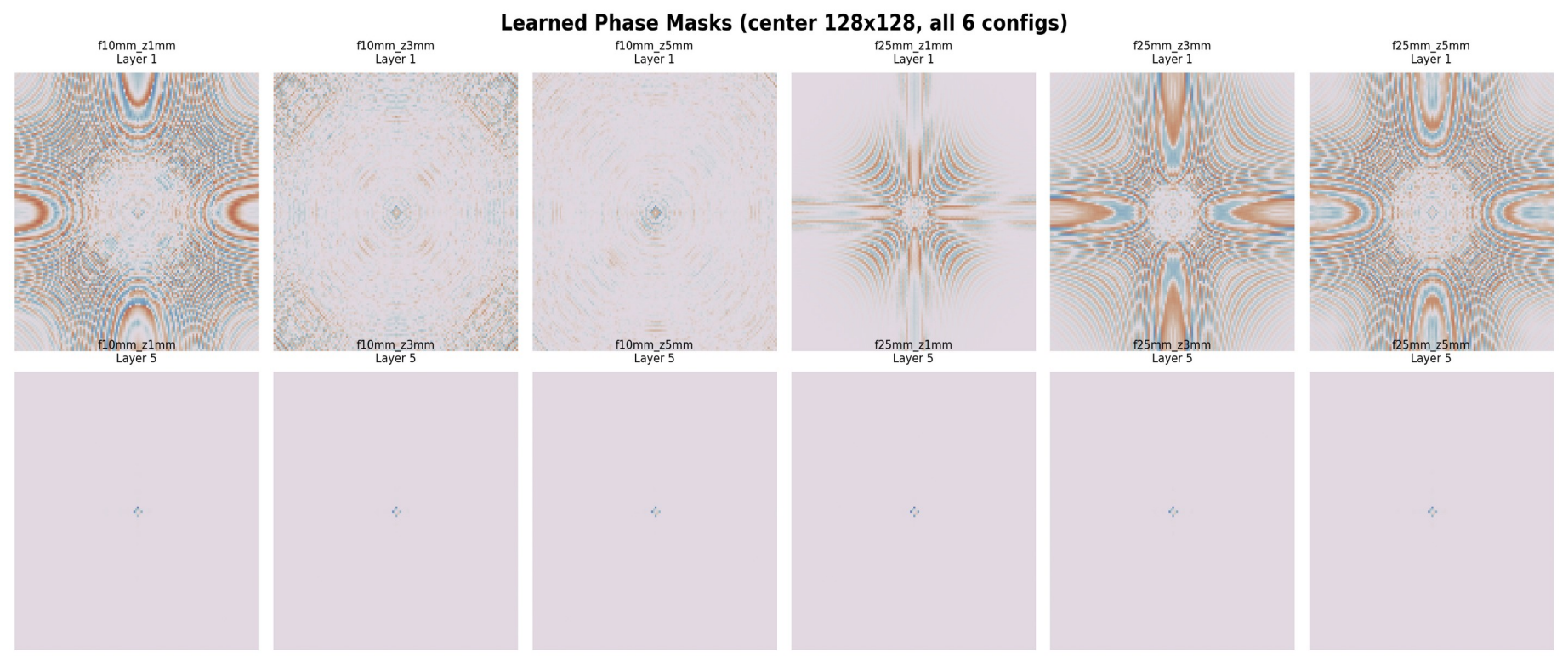


Throughput ≈ 1.0 (모든 config 에서 에너지 보존). CO 는 모두 baseline(0.268) 미달 — static mask 의 근본적 한계 .

물리적 해석 : Diffraction Regime



학습된 Phase Masks (Layer 1 & 5)



f=10mm masks: 강한 패턴 (학습이 defocus 보상 시도). f=25mm masks: 약한 패턴 (거의 flat → 기존 빔 유지).

결론 및 시사점

1. f-z 딜레마 확인

f↓: 회절 mixing 강하지만 defocus 로 baseline 미달 (최대 -0.324)
f↑: 안정적이지만 회절 mixing 약해 실질 효과 없음 (+0.010)

2. Spatial D2NN 대비 열위

Spatial D2NN: PIB +0.240 (mode conversion) vs FD2NN: PIB +0.010 (Fourier 위상 보정 한계)

3. 근본 원인 : Static mask 로 랜덤 위상 보정 불가

Wiener filter 최적해 H_opt → 0. PSD 가 단순해도 개별 realization 의 θ(κ) 는 매번 랜덤 .

4. 향후 방향

Adaptive mask (CNN 기반 per-input mask 생성) 또는 Spatial D2NN 에 집중 .